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MAST20030: Differential Equations
Tutorial: Week 10

Question 1

In this question we are going to solve the 1D heat equation subject to inhomogeneous Neumann boundary conditions,

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad \partial_x u(0,t) = b, \quad \partial_x u(L,t) = b \quad \text{and} \quad u(x,0) = f(x), \tag{1}$$

where $\alpha, b \in \mathbb{R}_{\neq 0}$.

(a) Split the solution into steady-state and transient solutions, i.e., $u(x,t) = u_S(x) + u_T(x,t)$ where $u_S(x)$ satisfies the inhomogeneous boundary conditions. Find $u_S(x)$ and hence define a homogeneous IBVP for $u_T(x,t)$.

(b) Use separation of variables to find the general solution to $u_T(x,t)$.

(c) Use orthogonality of the eigenfunctions to write down a formula for the series coefficients in (b) and hence determine the solution to (1). What happens to the unknown constant that arises in the steady-state solution?

Question 2

In this question we are going to solve the 2D Laplace equation on a square plate subject to the following boundary conditions:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad u(0,y) = 0, \quad u(L,y) = 0, \quad u(x,0) = 0 \quad \text{and} \quad u(x,L) = \begin{cases} x, & 0 < x < L/2 \\ L - x, & L/2 < x < L \end{cases}. \tag{2}$$

(a) Use separation of variables to turn the PDE into multiple ODEs.

(b) Apply the first two (homogeneous Dirichlet) boundary conditions to the ODE governing $X(x)$ to find the eigenvalues/eigenfunctions.

(c) Solve the ODE governing $Y(y)$ for each eigenvalue.

(d) Use superposition to write down a series general solution that satisfies the PDE and first two boundary conditions in (2).

(e) Apply the last two boundary conditions in (2) to determine the series coefficients (using orthogonality) thereby finding the complete solution to (2).